



MELODEX

Power User Manual

Providers, resolver, playback gateway,
MCP, security and release architecture

Don't shuffle. Flow.

Version 0.2 release candidate



Local-first | Source-neutral | Optional AI

September 2026

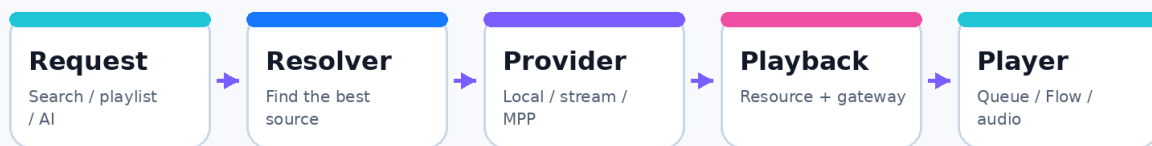
Power-user map

This manual focuses on the parts most listeners can ignore: source architecture, provider runtime, resolver behaviour, playback gateway, external control and release engineering.

DESIGN PRINCIPLE

Melodex aims for a clean core and a messy edge. Core understands normalized music and playback resources; providers absorb source-specific complexity.

Melodex architecture



Core owns:

playback | queueing | Flow | taste memory | resolver memory | AI/MCP control

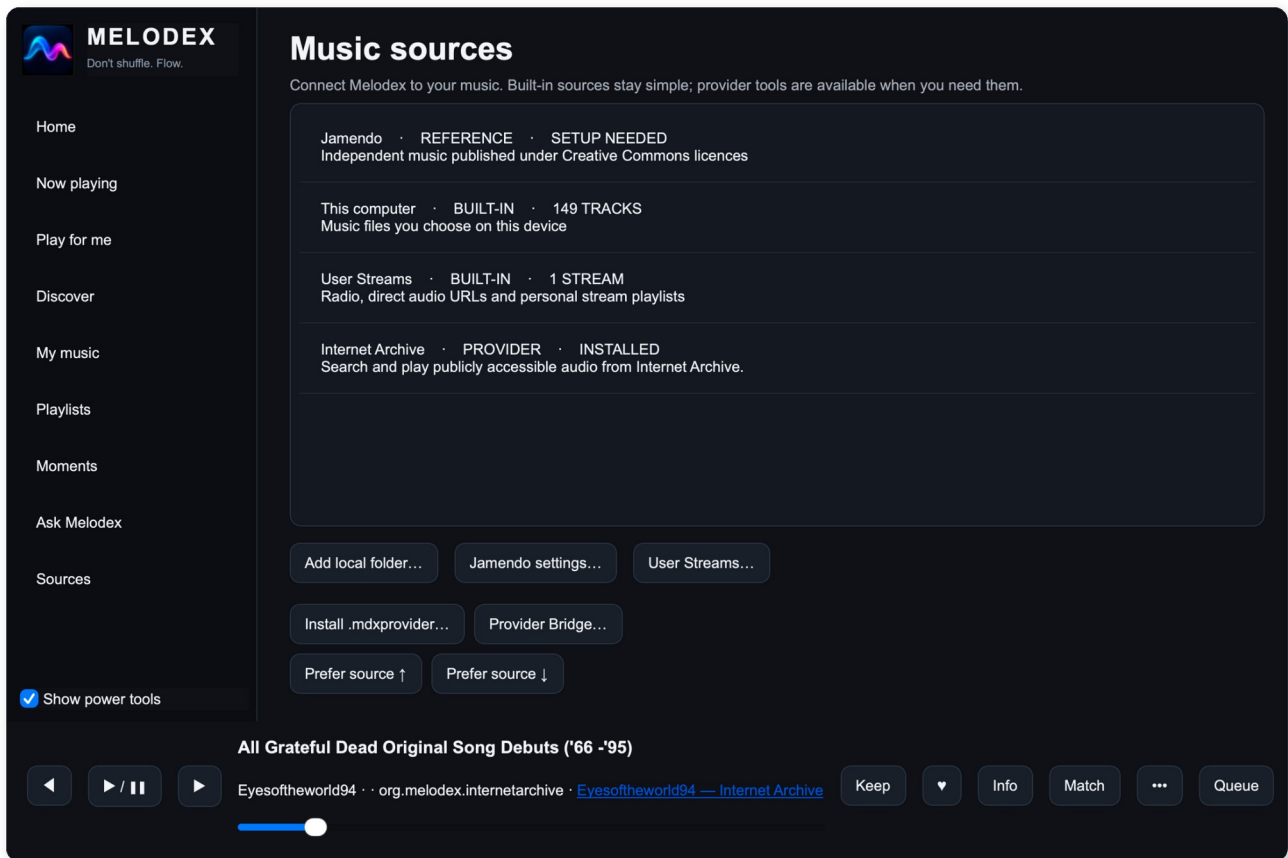
Providers own source-specific discovery and authorised playback access.

The request, resolver, provider and playback layers are deliberately separate.

- Core owns playback, queueing, Flow, taste memory and control surfaces.
- The Universal Resolver decides where a requested song should come from.
- Providers supply catalog metadata and authorised playback access.
- The Playback Gateway normalizes difficult-but-authorised network playback for Qt.

1. Sources, providers and trust

Built-in sources stay simple. Installable providers are executable software and should be treated accordingly.



Sources with power tools enabled - built-in sources, official providers and provider controls share one surface.

Control	What it means
This computer	Built-in indexed local files.
User Streams	Built-in user-supplied HTTP(S) radio/audio endpoints.
Jamendo	Reference provider using a developer client ID.
Internet Archive	Official optional MPP provider for publicly accessible Archive audio.
.mdxprovider	Third-party executable provider package running out-of-process.

TRUST MODEL

Out-of-process does not mean sandboxed. Third-party provider code still runs with the normal permissions of the operating-system user unless you add OS-level sandboxing.

2. Provider SDK 0.2 and MPP 1.0

The SDK keeps v0.1 providers compatible while adding the state needed for difficult-but-authorized web playback.

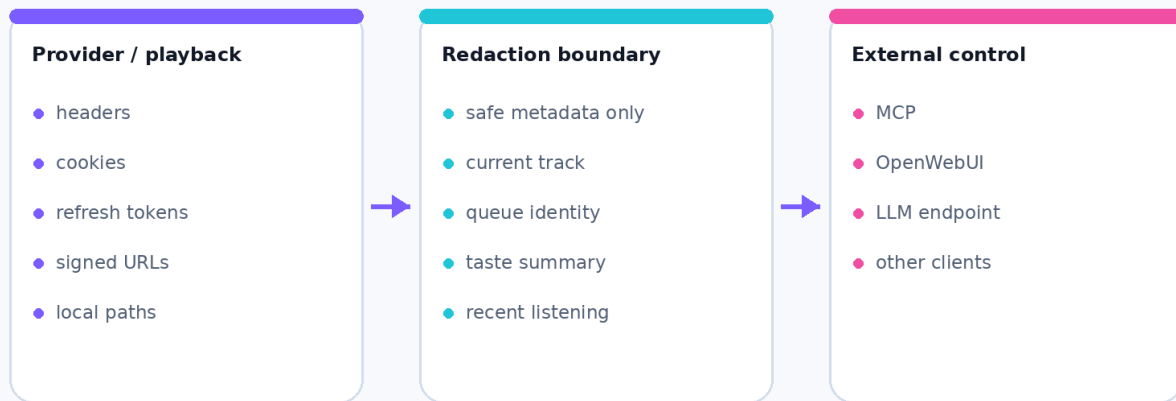
```
cd provider-sdk
python -m venv .venv
source .venv/bin/activate
python -m pip install -e '.[dev]'
melodex-provider init my-provider
melodex-provider validate my-provider
melodex-provider doctor my-provider
melodex-provider pack my-provider
```

- Clean providers can map JSON APIs directly to normalized objects.
- Messy providers may deal with HTML/XML, detail pages, redirects, sessions and temporary CDN URLs.
- Vendored Python dependencies can ship in a provider-local vendor/ directory.
- The public SDK excludes DRM circumvention, CAPTCHA bypass, anti-bot evasion and private credentials.

3. Playback resources and the local gateway

Playback state is richer than metadata. Headers, cookies, expiry and refresh state are internal playback material.

Security boundary: playback secrets stay inside Melodex



Only redacted control data crosses this boundary.

Sensitive playback fields remain on the inside of the redaction boundary.

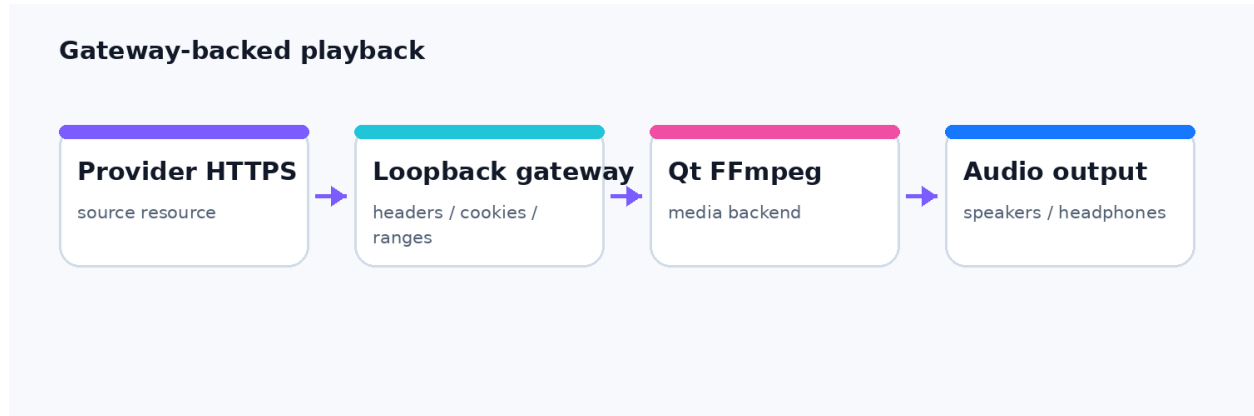
Control	What it means
url / stream_url	The provider-resolved media resource.
headers	Request headers needed by the resource.
cookies	Session state required for playback.
expires_at	Resource-expiry metadata.
refresh_token	Opaque refresh state for re-resolving an expired resource.
allowed_hosts	Provider-declared network-host boundary.
gateway_required	Force playback through Melodex's loopback gateway.

SECURITY HARDENING IN 0.2

Status, LLM and external-control responses redact private playback fields and ordinary listening history is not supposed to retain those secrets.

4. Why the gateway exists

Qt multimedia can behave differently across platforms. The gateway gives Melodex one controlled loopback path for stateful network playback.



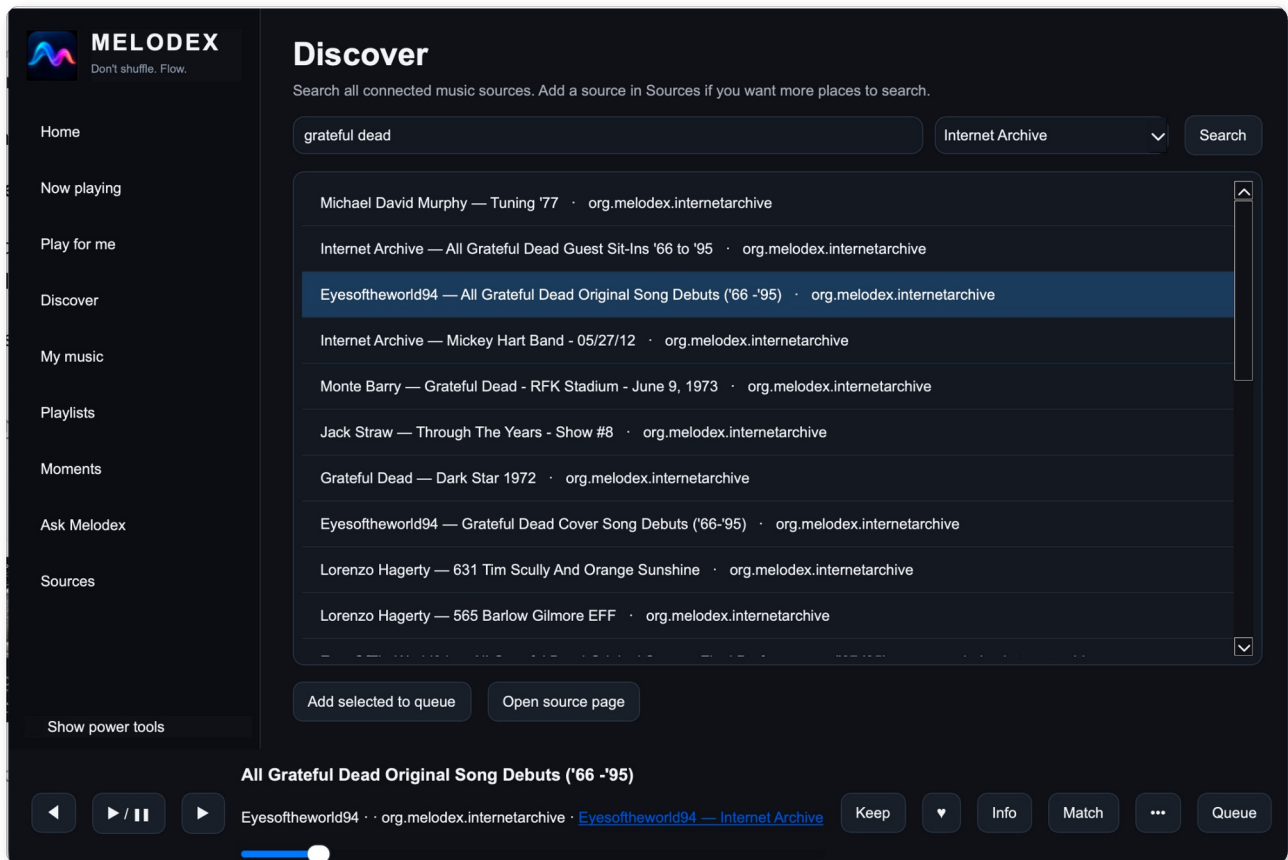
Provider HTTPS resource -> Melodex loopback gateway -> Qt FFmpeg -> audio output.

- Loopback-only listener.
- Unpredictable registration tokens.
- Byte-range support for media seeking/probing.
- Provider-declared host validation.
- No DRM or access-control bypass.

On macOS, the packaged v0.2 path uses the Qt FFmpeg backend so gateway-backed Internet Archive audio produces audible output reliably.

5. Internet Archive as a reference provider

The official Archive provider demonstrates the MPP v0.2 path on a real public source.



Internet Archive search results are item-level for fast discovery; file expansion happens later when playback is resolved.

```
PYTHONPATH=provider-sdk/src desktop/.venv/bin/python \
-m melodex_provider_sdk.cli doctor official-providers/internet-archive \
--query 'Grateful Dead'
```

- Uses public Advanced Search and item metadata.
- Preserves source page plus available rights/licence information.
- Uses public download endpoints only.
- Does not bypass login, lending, DRM or restricted-item protections.
- Startup time depends heavily on the selected Archive item and media size.

6. Universal Resolver and match memory

A request names the music. The resolver decides which connected source is an acceptable playable match.

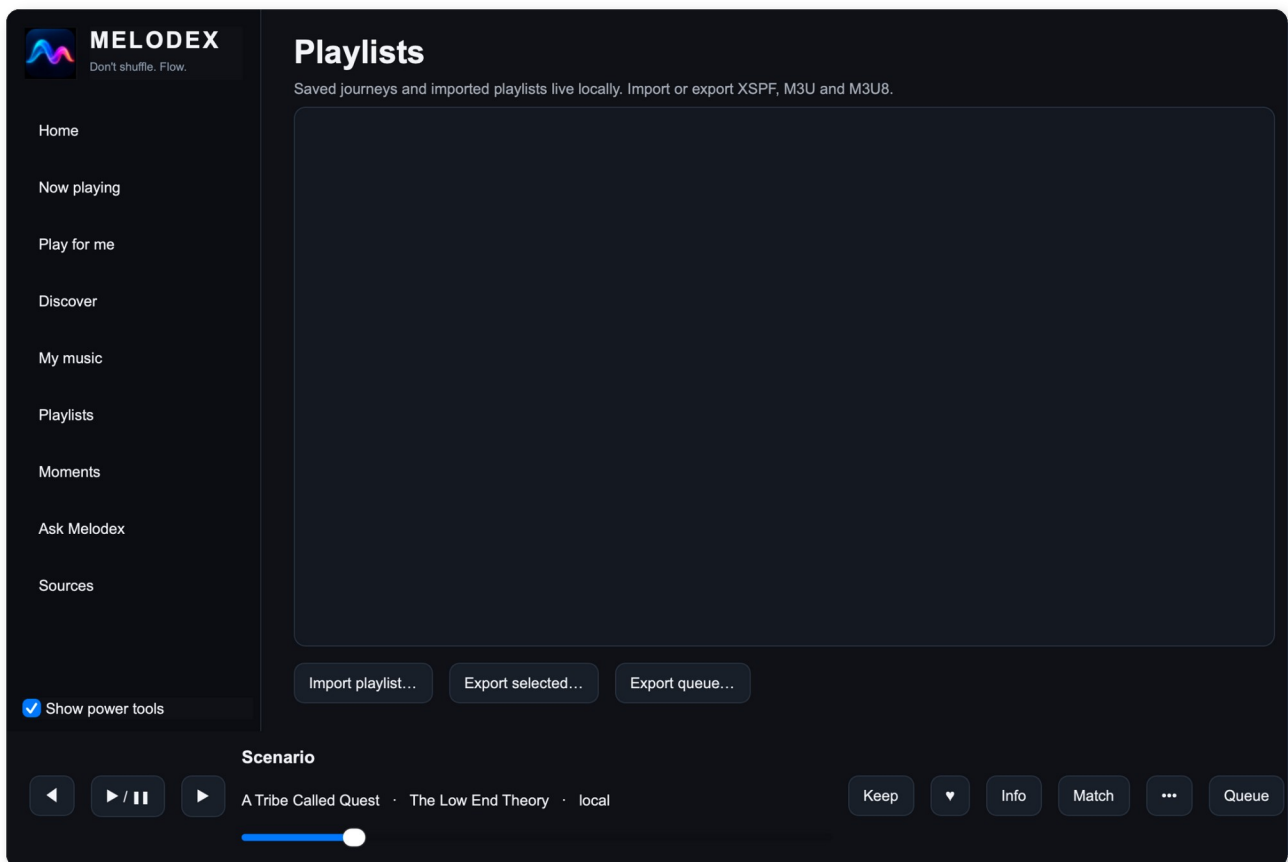
```
{"artist": "Artist", "title": "Track", "album": "Album"}
```

- Title similarity has the largest weight, followed by artist and album.
- Provider priority is a small tie-break rather than the main signal.
- Duration can refine a match when both sides provide it.
- Major variant mismatches such as live/remix/instrumental/karaoke/cover/acoustic are penalized.
- Candidates below the confidence threshold are not auto-played.

Control	What it means
Prefer	Remember one candidate for one requested song.
Wrong match	Block one candidate for one requested song.
Reset	Clear the song-specific resolver memory.

7. Playlist portability

XSPF, M3U and M3U8 can carry fixed locations or just enough metadata to let the resolver choose the source later.



Playlist interchange keeps the queue portable across different local libraries and provider combinations.

- Local paths can remain direct local entries.
- Direct HTTP(S) resources can remain fixed stream entries.
- Metadata-only entries can resolve on the listener's own machine.
- Melodex writes ignorable metadata so provider identity can survive round-trips where possible.

8. Rich metadata is independent of playback

A provider can supply the audio while Melodex independently enriches the track with public music knowledge and local lyrics.

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Now playing
Artwork, local lyrics, artist relationships and recording credits are enriched independently from the playback source.

What Would Biggie Do
Goldie Lookin Chain
Kings of Caerleon · 2013
Group | Newport | british · rock and indie · uk · welsh
✓ MusicBrainz match 99% · enrichment complete
[MusicBrainz recording](#) · [artist](#) · [release](#)

Goldie Lookin Chain
Welsh comedic rap group
Group · From Newport · Active from 2000

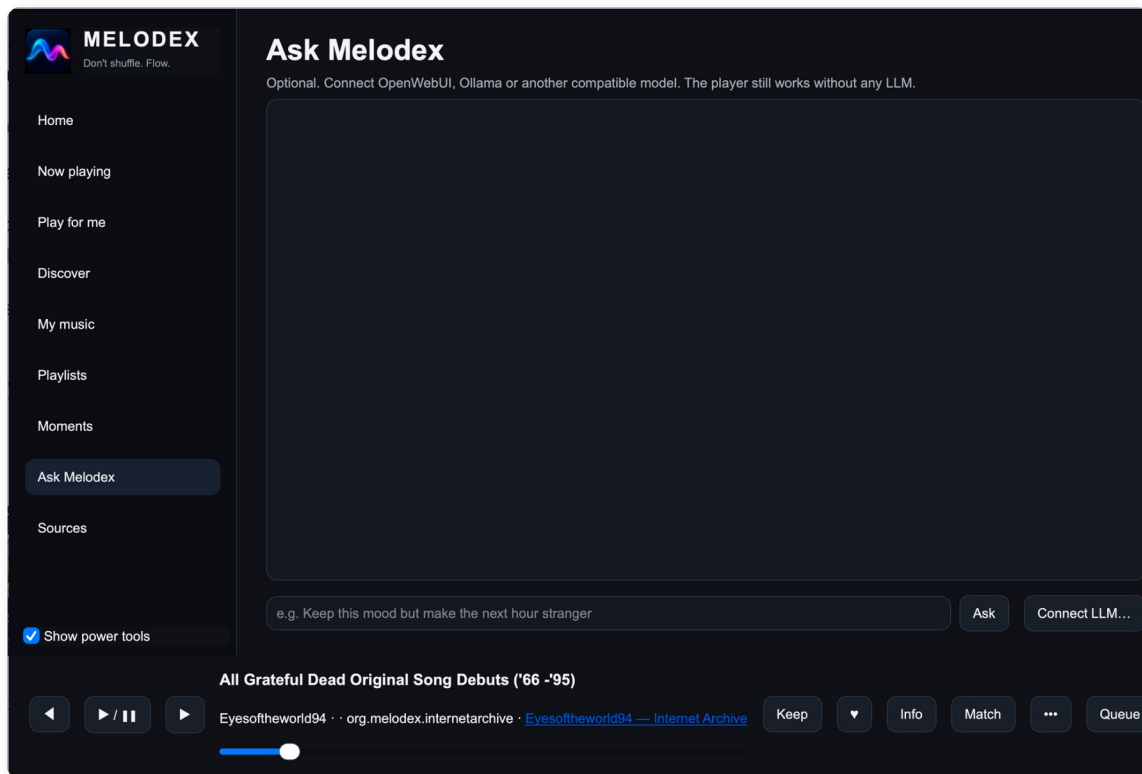
What Would Biggie Do_
Goldie Lookin Chain · Kings of Caerleon · local
Keep ♥ Info Match ... Queue

Rich Now Playing - playback source and metadata source are intentionally decoupled.

- Local tags and embedded artwork/lyrics.
- MusicBrainz recording, artist and release identity.
- Cover Art Archive artwork.
- Wikidata-linked Wikimedia Commons artist images with file-level attribution.
- Local .lrc and .txt lyrics sidecars.

9. Built-in LLM control and MCP

AI is a control layer, not a playback dependency. Melodex keeps Qt/playback inside the desktop process and exposes narrow authenticated control surfaces.



Ask Melodex - the built-in natural-language surface is optional.

REDACTED EXTERNAL STATUS

MCP and LLM control receive safe metadata rather than raw provider headers, cookies, refresh tokens, signed stream URLs or local paths.

- Private desktop bridge binds to loopback and uses a random bearer token.
- The MCP HTTP server uses a separate bearer token.
- MCP exposes status, search, resolve, play, queue, Flow, feedback and resolver-transparency tools.
- Do not expose an unauthenticated MCP server on a non-loopback interface.

```
python -m melodex.mcp_server --transport streamable-http --host 0.0.0.0 --port 8787
python -m melodex.mcp_server --show-token
```

10. Provider Bridge and Android

Provider Bridge is not MCP. It is the authenticated path used by the Android preview to search and play sources exposed by desktop Melodex.

```
music/provider -> Melodex desktop -> Provider Bridge (:8766) -> Android
```

- Use LAN mode only on a trusted network.
- Treat the bearer token like a password.
- Do not raw-port-forward the Bridge to the public internet.
- Android intentionally does not execute arbitrary downloaded provider code.

11. Local state and security review

Release engineering should assume that application-data files can contain private configuration even when LLM/control responses are redacted.

Control	What it means
macOS	~/Library/Application Support/Melodex/
Windows	%APPDATA%\Melodex\
Linux	\${XDG_DATA_HOME:-~/.local/share}/melodex/

- sources.json can contain provider configuration and resolver memory.
- The listening/taste database contains personal listening history.
- Bridge and MCP token files are credentials.
- Metadata/artwork cache may contain externally licensed content.

PUBLISHING RULE

Do not attach your application-data directory to a bug report without reviewing it first. Never paste provider credentials or bearer tokens into public issues.

12. Release candidate builds

v0.2 release candidates are built natively in GitHub Actions before the final tag is created.

Control	What it means
macOS Apple Silicon	DMG
macOS Intel	DMG
Windows x64	Setup EXE
Windows portable	ZIP
Linux x86_64	ApplImage
Android preview	APK + AAB
Final tag	Source ZIP plus tagged release assets

Unsigned preview builds may trigger operating-system warnings. Production distribution requires the relevant signing or notarisation identities.

Release gate:

- 1 Update all version files and release notes.
- 2 Run scripts/release_check.py.
- 3 Run desktop and Provider SDK test suites.
- 4 Build and smoke-test packaged desktop binaries.
- 5 Confirm external Python providers launch correctly from packaged builds.
- 6 Verify local playback and at least one gateway-backed provider.
- 7 Verify LLM/control redaction tests.
- 8 Build RC artifacts on GitHub Actions.
- 9 Update documentation and manuals.
- 10 Merge the release branch and tag v0.2.0.

13. Power-user quick reference

The boundaries below are the ones worth remembering when extending or deploying Melodex.

Control	What it means
.mdxprovider	Executable code. Trust the publisher.
Playback secrets	Internal state, not metadata.
Network hosts	Part of provider trust and gateway enforcement.
Resolver memory	Song-specific preference/blocklist.
Bridge token	Password for Android/bridge access.
MCP token	Password for external AI/tool access.
Local paths	Do not expose through external control surfaces.
Legal access	A provider protocol does not grant permission to access a source.

CORE IDEA

The resolver decides where a requested song should come from. Flow decides where the listening session should go next.

Don't shuffle. Flow.